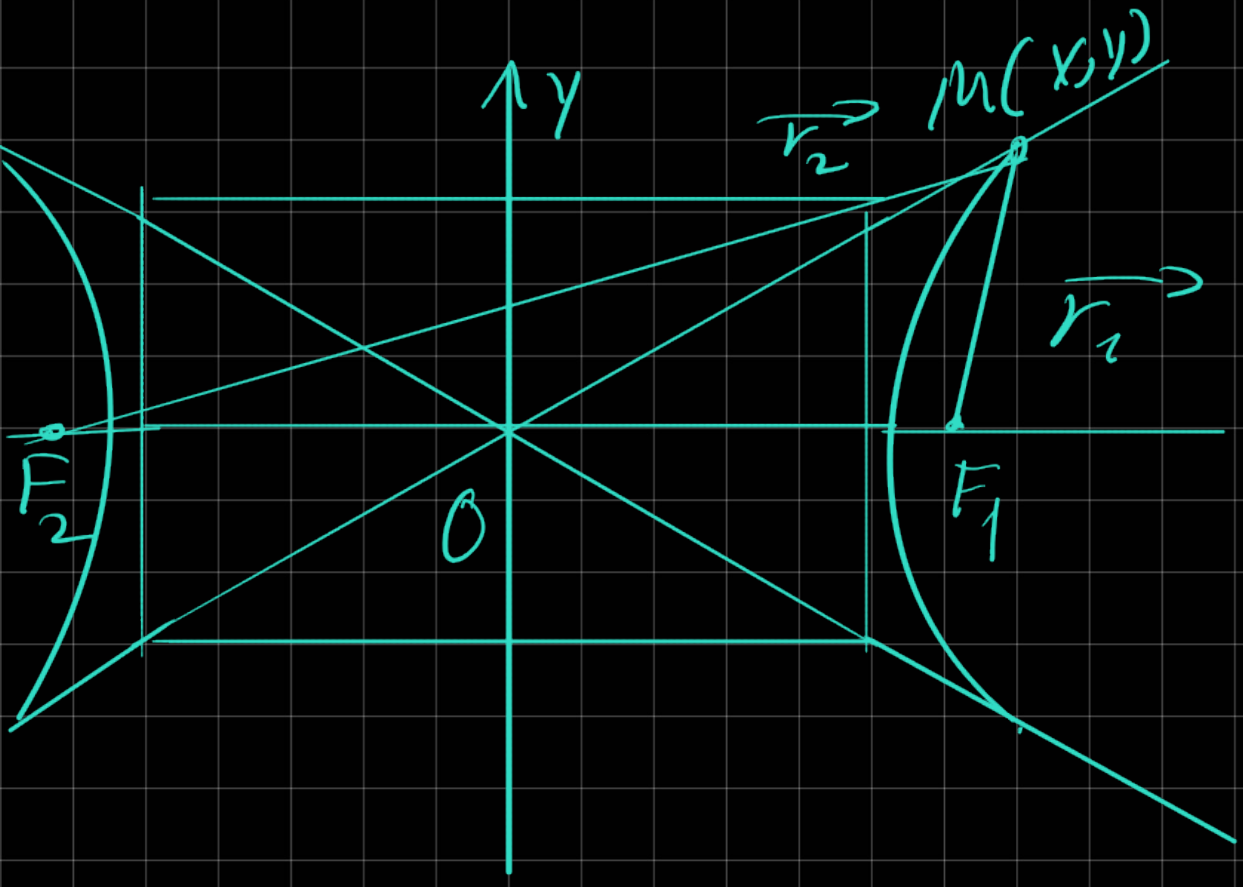


$$\{M \mid (|\overrightarrow{F_1M}| - |\overrightarrow{F_2M}| = 2a\}$$

$$|\vec{r} - \vec{r_1}| - |\vec{r} - \vec{r_2}| = \pm 2a$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Элементы гиперболы

1) O – центр
2) A, A', B, B'
3) $F_1(c_1; 0)$, $F_2(c_2; 0)$ – фокусы

4) $2c$ – фокусное расст.

$$c = \sqrt{b^2 + a^2}$$

5) $AA' = 2a$ — действ. ось

$BB' = 2b$ — мнимая ось

6) $e = \frac{c}{a} = \sqrt{\frac{b^2+a^2}{a^2}} = \sqrt{1 + \frac{b^2}{a^2}}$

$e > 1$

7) $x = \pm \frac{a}{e}$

8) $y = \pm \frac{b}{a} x$

9) $p = \frac{b^2}{a}$

10) $r_{1,2} = |a \pm e x|$

Парабола

$K\left(-\frac{p}{2}; y\right)$ $M(x;y)$ $F\left(\frac{p}{2}; 0\right)$

$\{M : |\vec{FM}| = d(M;\ell)\}$

$|\vec{r} - \vec{r}_1| = p + (\vec{r} - \vec{r}_1)\vec{e}$

Вывод в канон.

$|\vec{FM}| = |\vec{MK}|$

$r = d$; $d = \frac{p}{2} + x$

$r = \sqrt{\left(x - \frac{p}{2}\right)^2 + y^2}$

$\sqrt{\left(x - \frac{p}{2}\right)^2 + y^2} = \frac{p}{2} + x \quad (\uparrow 2)$

$x^2 - px + \frac{p^2}{4} + y^2 = \frac{p^2}{4} + px + x^2$

$y^2 = 2px$

$(x - x_0)^2 = 2p(y - y_0)$

$C_0(x_0; y_0)$

$x^2 = 2py$ $F\left(0; \frac{p}{2}\right)$ $y = -\frac{1}{2}p$

$x^2 = -2py$ $y = \frac{p}{2}$ $F\left(0; -\frac{p}{2}\right)$

$y^2 = -2px$ $F\left(-\frac{p}{2}; 0\right)$ $x = \frac{p}{2}$

Ф-ии пар.:

1) $O(0;0)$ верш. пар.
2) Ox – ось симм.
3) $x = -\frac{p}{2}$ – директр. пр.
4) $p > 0$ – фокальн. парам.
5) $\varepsilon = 1$

Преобразования координат



в XOy: $M(x;y)$

в XO_1y': $M(x';y')$

$$\begin{cases} x = x' + x_0 \\ y = y' + y_0 \end{cases} \qquad \begin{cases} x' = x - x_0 \\ y' = y - y_0 \end{cases}$$

Полярная система координат



$\rho = |\overrightarrow{OM}|$

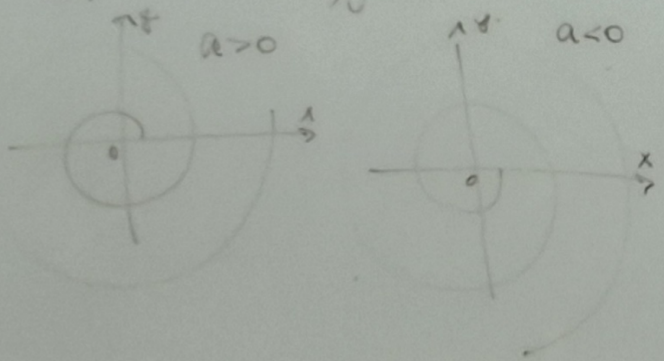
$0 \le \rho < +\infty$

$0 \le \varphi \le 2\pi$

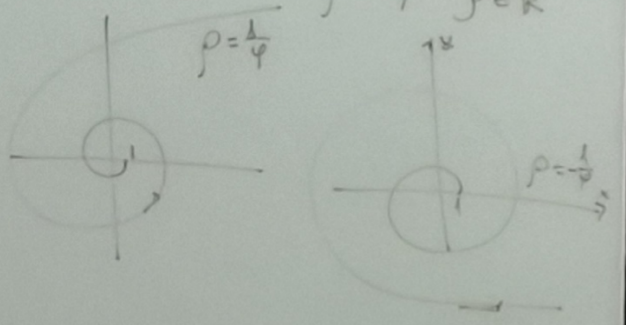
Спирали

1) Архимедова

$\rho = a\varphi \quad a < 1 < +\infty; \ \rho \in \mathbb{R}$



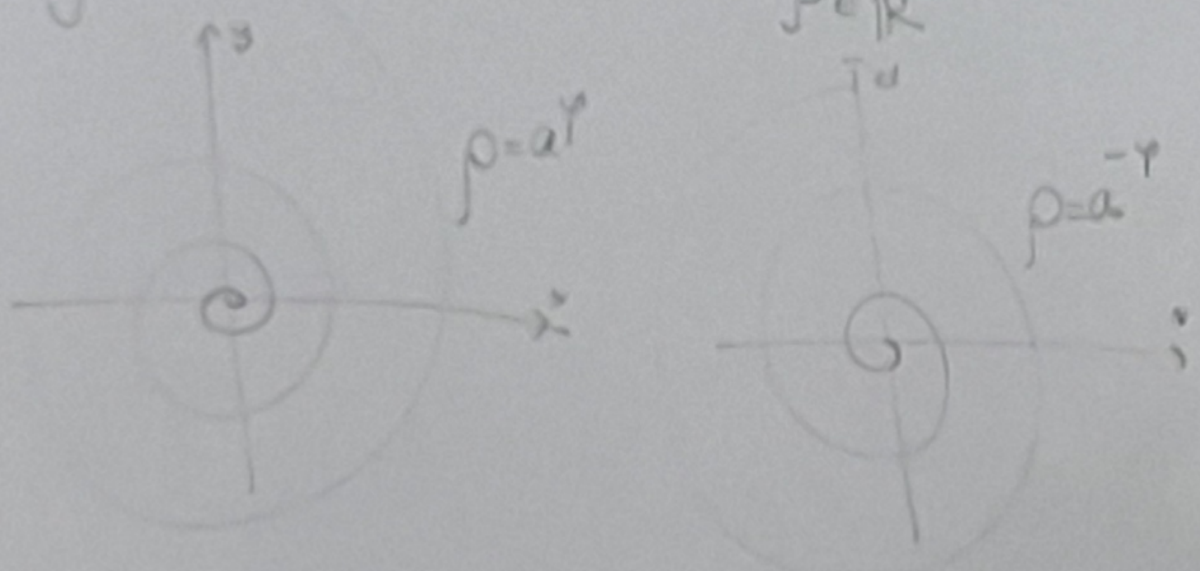
Гиперболическая сп. $\rho = \frac{a}{\varphi}$ $\varphi \in (0; \infty)$ $\rho \in \mathbb{R}$



Логарифмическая спираль

$\rho = a^{\varphi} \quad a > 0; \ a \neq 1 \quad \varphi \in [0; +\infty)$

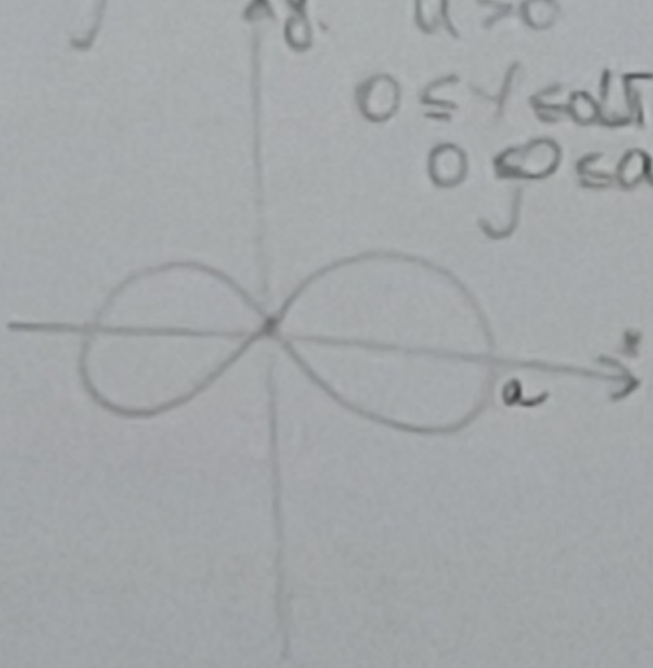
$\rho \in \mathbb{R}$



Розн

2) Кривые

$\rho = a\cos 2\varphi$

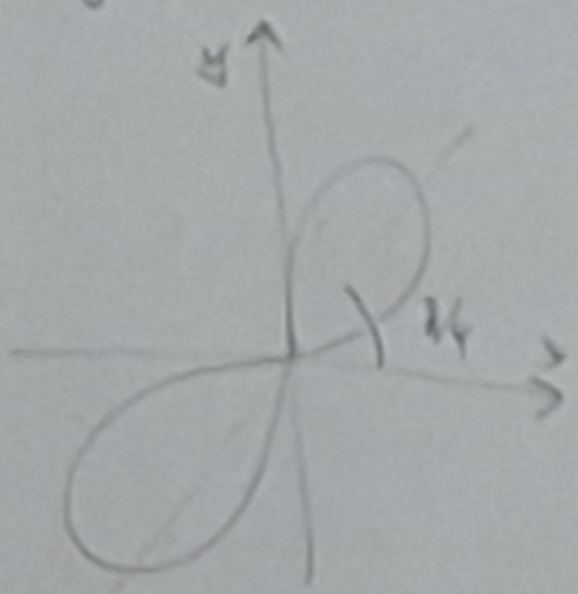


$a > 0$

$0 \le \varphi \le \frac{\pi}{4}$

$0 \le \rho \le a$

$\rho = a\sin 2\varphi$



ПРАКТИКА

Дз 3

$A_1(4, 7, 8)$ $A_2(-1, 13, 0)$ $A_3(2, 4, 9)$

$A_4(1, 8, 9$

1) пл-ти $A_1A_2A_3$

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

$$6x - 7y - 9z + 97 = 0$$

б) $\frac{x-4}{5} = \frac{y-7}{-6} = \frac{z-8}{8}$

2) $A_4N \parallel A_1A_2$

$$\frac{x-4}{5} = \frac{y-7}{-6} = \frac{z-8}{8}$$

$$S_1 = S_2 = (5; -6; 8)$$

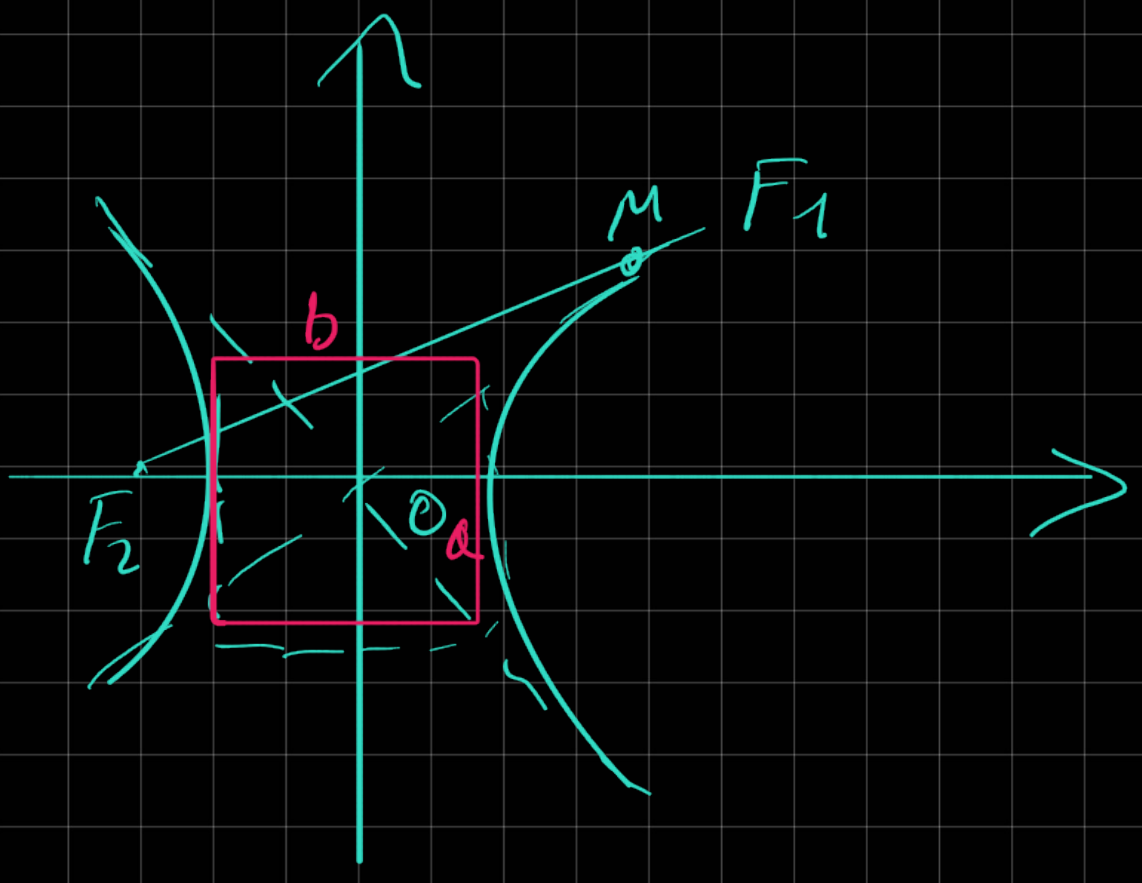
д) Синус угла между прямой A_1A_4 и $A_1A_2A_3$

$$|\cos(\vec{n}\,\hat{\vec{s}})| = \sin\varphi = \frac{|Am+Bn+Cp|}{\sqrt{A^2+B^2+C^2}\sqrt{m^2+n^2+p^2}}$$

$n_1 = (0, 0, 1)$

$n_2 = (6, -7, -9)$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



$r_1 = a + \varepsilon x$

$r_2 = -a + \varepsilon x$

$5x^2 - 4y^2 = 20$

$\frac{x^2}{4} - \frac{y^2}{5} = 1$

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$a = 2 \quad b = \sqrt{5}$

$A_1(a;0) \quad A_2(-a;0)$

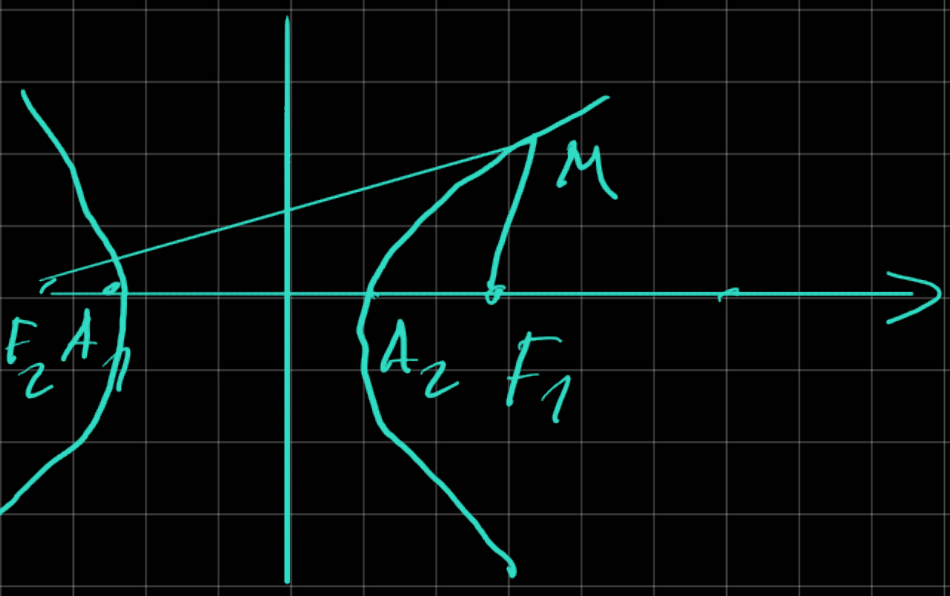
$2;0 \qquad -2;0$

$c = \sqrt{b^2 + (\sqrt{5})^2} = 3$

$y = \frac{1}{2}\sqrt{5(x^2 - 4)}$

$x = 3 \quad y = 2,5$

$x = 4 \quad y \approx 3,87$



$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$2C = 10$

$C = 5$

$b = 4$

$25 = a^2 + 16$

$a = 3$

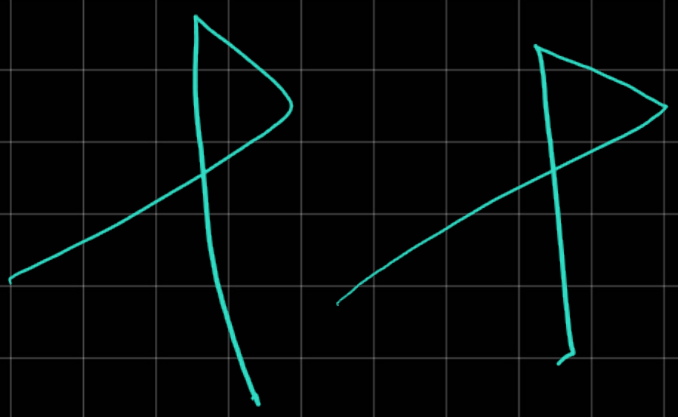
$$\frac{y^2}{16} - \frac{x^2}{9} = 1$$

$F_1(-2; 4)$ $F_2(12; 4)$

$$x_0 = \frac{(-2+12)}{2} = 5 \qquad y_0 = \frac{4+4}{2} = 4$$

$2c = 12 - (-2) = 14$

$c = 7$



$b^2 = 9$

$c^2 = a^2 + b^2$

$a^2 = 49 - 9 = 0$

$$\frac{(x-x_0)^2}{a^2} - \frac{(y-y_0)^2}{b^2} = 1$$

$$\frac{(x-5)^2}{48} - \frac{(y-4)^2}{9} = 1$$

$y^2 = 2px$

$p > 0$

$F\left(\frac{p}{2}; 0\right)$



$x^2 = 2py$

$(y-y_0)^2$

$$(x-x_0)^2 = 2p(y-y_0)$$
$$(x-x_0)^2 = -2p(y-y_0)$$

$$y^2 = 2px$$
$$x^2 = 4y$$
$$2p = 4$$

$$y = -2x^2 + 8x - 5$$
$$-2(x^2 - 4x + \tfrac{5}{2})$$
$$-2(x^2 - 4x + 4 - 4 + \tfrac{5}{2})$$
$$-2((x-2)^2 - \tfrac{3}{2})$$
$$-2(x-2)^2 + 3$$